

Comprehensive Supply Chain Management Framework for Utility-Scale Solar PV EPC Projects: A Case Study of the HELIOS 100 MW Project in Uzbekistan

Supply Chain Management Department Overview

Supply Chain Management (SCM) within the context of a utility-scale Solar Engineering, Procurement, and Construction (EPC) environment transcends the rudimentary administrative function of purchasing. It operates as the central nervous system of project execution, orchestrating a complex, globally distributed network of risk mitigation, cash-flow optimization, schedule safeguarding, and quality assurance. In the execution of the 100 MW HELIOS Solar PV project in Uzbekistan, the SCM function ensures that highly engineered electrical equipment, thousands of tons of bulk commodities, and specialized labor converge at an arid site in Central Asia precisely when the Primavera P6 construction schedule dictates. Rather than merely acquiring goods, SCM engineers the acquisition lifecycle to protect the EPC contractor's margin against currency fluctuations, commodity price volatility, geopolitical transit risks, and local regulatory bottlenecks.

The Procurement Plan serves as the foundational architecture for the entire SCM operation. Developed during the project's initiation phase, this document maps every material and service requirement against the baseline construction schedule. The plan dictates the procurement strategy for each package, establishing the critical dates for issuing the Request for Quotation (RFQ), receiving bids, conducting technical and commercial evaluations, issuing the Purchase Order (PO), conducting Factory Acceptance Tests (FAT), and achieving Ex-Works and site delivery milestones. The development of the Procurement Plan requires intense cross-functional collaboration. Engineering provides the Material Take-Offs (MTO) and technical specifications; Planning integrates lead times into the schedule; and Finance outlines cash-flow constraints and Letter of Credit (LC) issuance timelines. Once drafted, the Project Director and corporate SCM leadership approve the plan, establishing it as the definitive baseline against which SCM performance is measured.

A nuanced understanding of the project's acquisition landscape requires distinguishing between equipment procurement, bulk material procurement, and subcontract procurement. Equipment procurement focuses on high-value, highly engineered, long-lead items such as power transformers, Gas Insulated Switchgear (GIS), and solar modules. These items demand rigorous vendor pre-qualification, extensive design reviews, stage-wise factory inspections, and complex international logistics strategies. Bulk material procurement involves volume-based commodities such as structural steel, low-voltage cables, and civil construction

materials. The challenge here lies in managing dynamic quantities derived from evolving MTOs, minimizing wastage, and orchestrating phased deliveries to prevent site congestion. Subcontract procurement shifts the focus from physical goods to services and labor risk. Subcontracts cover civil works, mechanical erection, and electrical stringing, requiring intricate legal agreements that flow down the prime contract's Liquidated Damages (LDs), warranty obligations, and performance guarantees to the local or regional subcontractor. The day-to-day role of the SCM Manager involves navigating these intersecting disciplines. The manager functions as an executive expeditor, a chief negotiator, and a crisis diplomat. Daily activities include reviewing Technical Bid Evaluations (TBE) and Commercial Bid Evaluations (CBE) for critical path items, negotiating LC terms with international vendors, liaising with customs brokers to resolve border delays at the Chinese-Kazakh frontier, and arbitrating scope disputes with site subcontractors. The SCM Manager continuously scans the macroeconomic horizon for logistical disruptions, adjusting the Procurement Plan dynamically to shield the project's critical path.

Long-Lead Equipment Procurement

The procurement of long-lead equipment permanently governs the project's critical path. For the HELIOS 100 MW Solar PV project, which integrates a 33/220kV Grid Substation (GSS), several bespoke items dictate the baseline schedule. The engineering, manufacturing, and transit times for these components require procurement actions to commence immediately upon project award.

Equipment Type	Typical Lead Time (PO to Delivery)	Critical Path Rationale	Key Technical Parameters	Common Supplier Origins
220kV Power Transformer	24 to 30 Weeks	Heaviest, most complex bespoke item. Plant cannot be energized without it.	Vector group, impedance voltage, ONAN/ONAF cooling, short-circuit withstand capability.	China, India, Turkey, Europe
220kV Gas Insulated Switchgear	20 to 24 Weeks	Requires clean-room assembly, highly precise SF6 gas compartment	Rated voltage, short-circuit breaking current (e.g., 40kA/3s), IP rating (IP67/IP68),	China, Europe, South Korea

		manufacturing.	AFLR classification ¹ .	
Solar PV Modules	10 to 14 Weeks	Enormous volume dictates schedule. 100MW requires ~3,000+ containers.	Wp rating, bifaciality factor, degradation rates, temperature coefficients.	China
Inverters (Central/String)	12 to 16 Weeks	Dependent on global semiconductor supply chains and grid-code compliance firmware.	Max efficiency, MPPT voltage range, total harmonic distortion (THD).	China, Germany, Spain
33kV Ring Main Units (RMU)	12 to 14 Weeks	Critical aggregation points for the DC-to-AC conversion blocks.	36kV rating, SF6 temperature-compensated gauge, internal arc classification ³ .	China, India, Europe

The Request for Quotation (RFQ) process for these major packages is highly structured. An RFQ package contains the Commercial Terms and Conditions, the Technical Specification, Project-Specific Data Sheets, the Project Quality Plan, and the required delivery schedule. Bidders submit their proposals utilizing a strict two-envelope system to prevent commercial bias during the technical review.

The bid evaluation process begins with the Technical Bid Evaluation (TBE), conducted exclusively by the engineering department. The TBE strips away pricing to assess whether the proposed equipment meets the project specifications, identifying deviations and raising technical queries. Once the TBE is approved and a bidder is deemed technically acceptable, the SCM department opens the commercial envelope to perform the Commercial Bid Evaluation (CBE). The CBE normalizes prices by accounting for variable Incoterms (e.g., adjusting FOB prices to DDP Site prices), payment terms, energy efficiency capitalizations (such as penalizing a transformer design with higher no-load losses over a 25-year lifecycle), and warranty costs, establishing the true Total Cost of Ownership (TCO).

A Purchase Order represents the legally binding culmination of this process. It explicitly details the scope of supply, the agreed technical data sheets, delivery Incoterms (often CIP Tashkent or DDP Site), the delivery schedule, and rigorous penalty clauses for delayed delivery known as Liquidated Damages (typically 0.5% per week, capped at 10%). Payment terms for international equipment typically rely on irrevocable Letters of Credit (LCs), structured around strict milestones: a 10-20% advance payment against an Advance Payment Bank Guarantee (APBG), a 70-80% major disbursement at sight against shipping documents (Bill of Lading, Commercial Invoice, Packing List, and FAT certificates), and a final 10% retention release upon successful site commissioning⁵.

The Factory Acceptance Test (FAT) serves as the final quality gate before dispatch. Representatives from the EPC contractor, the project owner, and often a third-party inspector travel to the manufacturing facility. For a 220kV transformer, the FAT involves routine tests such as winding resistance measurement, voltage ratio checks, and dielectric tests, alongside type tests if specified. The successful completion of these tests generates the FAT Report, an essential document required to trigger the dispatch clearance, release the equipment to the logistics provider, and authorize the corresponding LC payment milestone.

Balance-of-Plant and Bulk Material Procurement

Balance-of-Plant (BOP) and bulk materials constitute the physical connective tissue of the solar facility. For a 100 MW project, the required volumes are colossal, and procurement efficiency here dictates the project's margin integrity.

Bulk Material Category	Typical 100 MW Project Quantity	Unit of Measure
Solar PV Modules (e.g., 600Wp)	~166,000 to 170,000	Pieces / Modules
DC Cable (4 sq mm / 6 sq mm)	1,200 to 1,500	Kilometers
AC Low Voltage Power Cables	40 to 60	Kilometers
33kV Medium Voltage XLPE Cable	25 to 35	Kilometers
Module Mounting Structure (MMS) Steel	3,000 to 3,500	Metric Tons
Earthing Materials (GI)	50 to 70	Metric Tons

Strips/Cu Rods)		
Civil Materials (Cement, Rebar)	Variable by terrain	MT / Cubic Meters

The conversion of Material Take-Offs (MTO) generated by engineering into executable purchase quantities requires strategic calculation. Engineering provides the net required quantities based on the 3D model or 2D layouts. SCM and construction planning must apply mathematical wastage factors. Typically, a 2% to 3% wastage factor is applied for structural steel, and up to 5% for cables to account for cutting waste, terrain undulations, trenching deviations, and termination lengths. Buffer stocks are strategically added to critical path commodities to prevent work stoppages caused by minor design changes or unavoidable site damage.

Bulk cable procurement demands highly specialized optimization techniques known as cable drum scheduling. Standard cable drums hold specific lengths (e.g., 500 meters or 1,000 meters depending on the cable diameter and drum weight limits). The objective is to map the required cable pulls from the engineering layouts against standard drum lengths to minimize the number of straight-through joints and scrap pieces. Mathematical optimization algorithms are frequently utilized to match specific circuit lengths to specific drums, generating a precise cutting schedule that the site execution team must strictly follow⁷. This reduces electrical resistance vulnerabilities at joint points and prevents the accumulation of unusable 30-meter offcuts.

Managing material specifications and the Approved Vendor List (AVL) ensures quality control across high-volume commodities. The AVL is a pre-negotiated roster of manufacturers whose technical capabilities, financial stability, and quality management systems have been vetted and approved by the project owner. SCM must ensure that all bulk materials are sourced exclusively from the AVL, preventing the introduction of sub-standard steel or non-compliant cables that could jeopardize the facility's 25-year operational lifespan.

Subcontract Procurement

Subcontract procurement transfers the execution risk of civil works, mechanical erection, and electrical installation from the prime EPC contractor to specialized local or regional entities. Typical subcontracts for a 100 MW solar EPC project include site grading, perimeter fencing, foundation piling, MMS and module installation, trenching and cable laying, and the highly specialized erection and testing of the 33/220kV Grid Substation.

A subcontract agreement is a complex legal instrument designed to protect the prime EPC contractor by mirroring the obligations of the main contract. Key clauses define the exact Scope of Work, tying it to specific "Approved for Construction" (AFC) engineering drawings and a firm Bill of Quantities (BOQ). Payment terms are almost exclusively milestone-based or tied to monthly progressive measurements, commonly referred to as Running Account (RA) bills. Liquidated Damages provisions are strictly enforced to penalize schedule overruns,

ensuring the subcontractor shares the financial pain of delays.

The most critical elements of these contracts are the back-to-back provisions. These clauses ensure that any risk, penalty, or technical obligation imposed on the EPC contractor by the ultimate project owner flows down directly to the subcontractor for their respective scope. The "pay-when-paid" clause is a hallmark of back-to-back structuring; it legally permits the EPC to withhold payment to the subcontractor until the EPC receives the corresponding funds from the project owner, thereby shielding the EPC's corporate cash flow.

The subcontract bid process mirrors equipment procurement but places heavier emphasis on the bidder's localized execution capacity and physical footprint. Subcontractors are evaluated on their mobilization speed, ownership of specialized equipment (such as hydraulic pile drivers or heavy excavators), available manpower histograms, and historical safety and environmental compliance records in Central Asia.

A critical distinction exists between a Subcontract Agreement and a Work Order. A Subcontract Agreement is a comprehensive, long-term contract governing complex scopes with high risk (e.g., total mechanical erection), containing extensive legal frameworks for dispute resolution, indemnification, performance bonds, and insurance requirements. A Work Order is a shorter, transactional document used for low-risk, short-duration tasks (e.g., hiring local security, temporary fencing, or emergency water-bowser services). The Work Order focuses on immediate deliverables and unit rates, bypassing the heavy legal architecture required for major works.

Expediting

Expediting is the proactive management of the supply chain timeline, ensuring that vendors adhere to the contractual delivery schedule. The expeditor functions as the project's early warning system. Day-to-day, the expeditor tracks the progress of hundreds of POs, routinely communicating with factory production managers across global time zones to verify that manufacturing milestones match the baseline schedule.

The expediting process follows a distinct lifecycle. It begins with tracking the Order Acknowledgement, ensuring the vendor has legally accepted the PO without unapproved exceptions. The next phase monitors Drawing Submission, pressing the vendor to submit engineering documents and simultaneously pushing the internal EPC engineering team to return them with approval, as manufacturing cannot commence without AFC drawings. Once approved, the expeditor monitors Production Start, confirming raw material availability (such as copper for cables or electrical steel for transformers) and manufacturing progress. As completion nears, the focus shifts to Inspection Readiness, coordinating the FAT dates with the quality department. Finally, the expeditor manages Dispatch and Delivery, coordinating with logistics providers to secure vessels, rail wagons, or truck trailers.

The expeditor relies on a comprehensive Expediting Report, updated weekly. This report functions as the single source of truth for procurement status, detailing the PO number, vendor name, item description, baseline delivery date, forecasted delivery date, current manufacturing status, identified bottlenecks, and immediate action plans. The expeditor acts as the vital interface between the factory floor, the logistics team (to book freight), and the site planning team (to adjust construction schedules based on accurate delivery forecasts).

Common causes of equipment delivery delays include global raw material shortages, failed quality inspections requiring rework, and a lack of available shipping containers or rail wagons. Expeditors manage these delays by demanding recovery schedules from the vendor, escalating issues to senior management, authorizing premium freight (e.g., air freighting critical electronic components), or suggesting construction sequence changes to the site team to allow civil works to continue around the delayed item.

Logistics & Customs — Uzbekistan Context

The logistical transit of heavy, high-value equipment into Uzbekistan—a doubly landlocked country—presents immense complexities and requires profound regional expertise.

The import process begins well before the cargo arrives at the border, utilizing the Unified Electronic Information System of Foreign Trade Operations (EEISVO). By law, the foreign trade contract must be registered in the EEISVO portal to obtain an Identification Number (IDN)¹⁰.

This IDN is mandatory; without it, customs clearance cannot proceed, and local banks will not authorize the transfer of foreign currency to pay the international vendor¹². Once the cargo arrives, customs clearance utilizes a Single Window electronic system (often interacting with ASYCUDA World), where the Cargo Customs Declaration (GTD) is submitted digitally by a licensed local customs broker¹⁴.

The documentation required for import clearance is rigorous. It includes the Commercial Invoice, Packing List, Transport Documents (CMR for road transport, SMGS for rail), Certificate of Origin (critical for claiming preferential tariffs under CIS agreements), and specific Certificates of Conformity¹⁷. Uzbekistan maintains its own mandatory certification system, known as GOST-UZ, managed by the Uzstandard agency. High-risk electrical equipment such as power transformers, switchgear, and inverters must obtain a GOST-UZ Certificate of Conformity (CoC) prior to customs release, requiring either single-batch certification based on foreign test reports or full serial certification valid for 1-3 years¹⁹.

The Harmonized System (HS) code landscape for solar equipment in Uzbekistan heavily influences project economics. To promote the energy transition, Uzbekistan applies a 0% import customs duty on solar panels and PV modules (HS Code 8541.40). However, a standard 12% Value Added Tax (VAT) is still levied on the CIF (Cost, Insurance, and Freight) value plus any applicable customs fees¹⁵. Power transformers (220kV) are classified under HS Code 8504.23.10, and Gas Insulated Switchgear falls under HS 8535.90.30²³.

Given the reliance on Chinese-manufactured equipment, transport routes are heavily dependent on the Sino-Kazakh border crossings. The primary road freight route exits China via the Khorgos port, transiting through Almaty and Shymkent in Kazakhstan, before crossing into Uzbekistan and proceeding to Tashkent—a journey of roughly 5,000 to 7,000 kilometers²⁵. Rail freight typically utilizes the Alashankou-Dostyk or Khorgos-Altynkol corridors. Because China utilizes standard gauge rail (1435 mm) while Kazakhstan and Uzbekistan utilize Russian broad gauge (1520 mm), all rail cargo must undergo transshipment or bogie exchange at the border²⁶. An alternative route involves sea freight to Bandar Abbas (Iran) followed by road transit through Turkmenistan, though this adds geopolitical complexity.

Abnormal load logistics require specialized engineering. A 220kV power transformer can weigh

upwards of 100 to 150 metric tons. Transporting this requires multi-axle hydraulic modular trailers to distribute the load and prevent structural damage to road infrastructure²⁸. In Uzbekistan, such transport necessitates special permits from the Committee for Roads (Uzavtoyul), requiring detailed route surveys to bypass low bridges, reinforce culverts, and navigate tight urban intersections³⁰.

The Last-Mile delivery plan bridges the gap between the main highway or railhead and the remote solar site. This involves coordinating escort vehicles, securing local traffic police approvals, and ensuring the site access roads are sufficiently compacted to bear the weight of the hydraulic trailers.

Upon arrival, the transport damage inspection process is critical. Due to the high risk of internal damage caused by train shunting, pothole impacts, or crane drops, power transformers are equipped with 3-axis impact recorders (ShockLog or similar devices) monitoring the X, Y, and Z axes for forces between 0g and 10g²⁹. If the recorder indicates that the transformer experienced a shock exceeding the manufacturer's safety limit (typically 4g), the equipment cannot be energized until exhaustive internal diagnostics (such as Sweep Frequency Response Analysis) are conducted, and a claim is initiated under the Institute Cargo Clauses (A) marine transit insurance³¹.

Site Stores Management

Site stores management transforms the chaotic influx of thousands of components into an organized, traceable inventory system, safeguarding the project's physical assets from damage, theft, and degradation.

The process begins with the Material Receipt Note (MRN). When a truck arrives, the stores in-charge verifies the physical quantities against the packing list and the original PO. The resulting MRN is a critical financial document; it formally acknowledges receipt, transferring the risk of loss to the EPC and triggering the vendor's right to invoice for the delivery milestone. Material inspection on arrival involves both the stores team and the quality control (QC) department. The stores team conducts a visual inspection for transit damage, verifying quantities and part numbers. The QC engineer subsequently inspects the material against the approved technical data sheets, verifying Mill Test Certificates (MTC) for steel or routine test reports for electrical components. Materials that pass are moved into the usable inventory. For sensitive equipment, preservation protocols are enacted immediately—such as ensuring nitrogen pressure is maintained in transformers or storing inverters in dry, temperature-controlled environments.

Inventory management at the site relies on a First-In, First-Out (FIFO) methodology, tracked via an Enterprise Resource Planning (ERP) system (like SAP MM) or robust spreadsheet databases. The system tracks the exact location of materials (open yard, covered shed, secure container). Replenishment orders for bulk consumables are triggered by predefined Minimum Order Quantity (MOQ) and reorder point thresholds, ensuring construction is never halted due to a lack of fasteners, cable ties, or grounding lugs.

When the construction team requires material, they submit a Material Issue Slip. This document, signed by an authorized construction manager, details the required items and allocates them to a specific project cost code or WBS (Work Breakdown Structure) element.

This ensures that the financial system accurately tracks the cost of goods actually consumed in construction versus goods sitting in inventory.

If the initial receipt inspection reveals missing items, a Shortage Report is immediately raised. This is directed to the logistics team, the SCM expeditor, and the vendor. The expected response is an immediate investigation into the packing process and the expedited dispatch of the missing components at the vendor's expense. For materials that fail the QC inspection, a Non-Conformance Report (NCR) is generated. Rejected materials are physically segregated into a dedicated quarantine zone to prevent accidental use, pending repair, return, or scrapping instructions from the vendor.

At the project's conclusion, material reconciliation is conducted. This process compares the total quantities procured against the quantities theoretically required by the final As-Built drawings, plus the agreed wastage factors. Any unaccounted-for variance indicates either unrecorded site theft, excessive undocumented wastage by subcontractors, or engineering miscalculations, leading to financial back-charges against the responsible parties.

Real Document Templates — Full Detail

1. Procurement Plan Structure

Section	Content Description	Example Data (HELIOS 100MW)
Package ID	Unique identifier for the procurement package	HEL-SCM-EQ-001
Package Name	Description of the equipment/material	220kV Power Transformer 100MVA
Category	Equipment, Bulk, or Subcontract	Engineered Equipment
Budget Limit	Approved baseline budget (USD)	\$850,000
Eng. Inputs Date	Date engineering will provide AFC specs	15-Nov-2025
RFQ Issue Date	Planned date to release to market	20-Nov-2025
Bid Receive Date	Deadline for vendor submissions	10-Dec-2025

TBE/CBE Finish	Completion of technical and commercial evaluations	20-Dec-2025
PO Award Date	Contractually binding order date	28-Dec-2025
Lead Time	Contractual manufacturing and transit time	24 Weeks
FAT Date	Scheduled Factory Acceptance Test	05-Jun-2026
Site Delivery Date	Required On-Site Date (ROSD)	20-Jul-2026

2. RFQ Package Cover Sheet

Field	Detail
Project Name:	HELIOS 100 MW Solar PV + 33/220kV GSS
RFQ Reference:	RFQ/HEL/25/004 – 33kV Ring Main Units
Issue Date:	October 3, 2026
Closing Date:	October 24, 2026, 16:00 UZT
Buyer Contact:	FORGE Agent (Procurement Lead)
Scope of Supply:	Design, manufacture, testing, and delivery (DDP Site) of 15 nos. 33kV RMUs.
Enclosures:	1. Commercial Terms & Conditions (Annex A) 2. Technical Specification & Data Sheet (Annex B)

	3. Project Quality Plan (Annex C) 4. Supplier Code of Conduct (Annex D)
Submission Rule:	Two-envelope system. Envelope 1: Unpriced Technical Bid. Envelope 2: Priced Commercial Bid. Any prices found in Envelope 1 will result in disqualification.

3. Technical Bid Evaluation (TBE) Matrix

Technical Parameter	Project Specification	Vendor A (Brand X)	Vendor B (Brand Y)	Vendor C (Brand Z)
Equipment	33kV RMU (Non-Extensible)	Offered	Offered	Offered
Standard	IEC 62271-200 AFLR	Complies	Complies	Deviation (Non-AFLR)
Rated Voltage	36 kV	36 kV	36 kV	36 kV
Short Circuit Rating	25kA for 3 sec	25kA for 3 sec	21kA for 3 sec	25kA for 3 sec
Gas Pressure Gauge	Temp-compensated, per IEC	Yes	Yes	No
Type Test Report	Required (< 5 years old)	Submitted	Submitted	Missing
Status / Conclusion	N/A	Technically Acceptable	Rejected (SC rating low)	Rejected (No Type Test)

4. Commercial Bid Evaluation (CBE) Format

Commercial Parameter	Vendor A (Technically Cleared)	Vendor D (Technically Cleared)
Base Price (Ex-Works)	\$225,000	\$210,000
Freight to Tashkent (CIP)	\$18,000	\$35,000
Total Delivered Price	\$243,000	\$245,000
Payment Terms	10% Adv, 80% LC, 10% Ret	20% Adv, 80% against dispatch
Financial Impact (NPV/Cashflow)	Favorable (LC defers cash outflow)	Unfavorable (High advance)
Delivery Schedule	12 Weeks (Meets Schedule)	16 Weeks (Delays Critical Path)
Liquidated Damages	Accepted 0.5% per week	Capped at 5% total (Deviation)
Recommendation	AWARD TO VENDOR A	Backup

5. Purchase Order Key Clauses

Section	Clause Description
PO Number & Date	PO-HEL-25-112
Vendor Details	ABC Heavy Industries, Guangzhou, China
Scope of Supply	1 No. 220kV/33kV 100MVA Power Transformer as per Spec TS-001.
Total PO Value	\$850,000 USD (CIP Tashkent via Khorgos port)

Payment Terms	1. 15% Advance against APBG. 2. 75% via Irrevocable LC at sight against shipping docs and FAT. 3. 10% upon successful site energization or 12 months from delivery.
Delivery Date	20-Jul-2026. Time is the essence of the contract.
Liquidated Damages	0.5% of PO value per week of delay, up to a maximum of 10%.
Warranty	24 months from commissioning or 30 months from delivery, backed by a 10% Performance Bank Guarantee (PBG).
Inspection & FAT	14 days advance notice required. EPC reserves right to witness all tests.
Transit Monitoring	Vendor must install 3-axis Impact Recorder (0-10g range).

6. Expediting Report (Weekly)

PO Number	Vendor	Item Description	Baseline Delivery	Forecast Delivery	Current Status	Blocker / Risk	Action Required
PO-112	ABC Ind.	220kV Transformer	20-Jul-26	27-Jul-26	Core assembly	Copper wire delayed	Escalate to ABC VP for 3-shift ops
PO-115	XYZ	100MW	15-May	15-May	Cell	None	Book

	Solar	PV Module s	-26	-26	stringin g		rail wagons at Alashan kou
PO-120	DEF Steel	MMS Structur es	10-Apr- 26	24-Apr- 26	Galvani zing	QA failed batch 2	Rework ordered . Deduct LDs.

7. Material Receipt Note (MRN)

Field	Detail
MRN Number:	MRN-HEL-26-405
Date of Receipt:	22-Jul-2026
Supplier:	Global Cable Co.
PO Number:	PO-HEL-25-202
Transporter & Waybill:	UZ-Trans Logistics, CMR #88392
Item Description:	1x500 Sq.mm 33kV XLPE Cable
Quantity Invoiced:	15,000 meters (15 drums)
Quantity Received:	15,000 meters
Visual Condition:	14 drums OK. 1 drum lagging damaged.
Stores Signature:	<i>[Signed VAULT Agent]</i>
QC Clearance:	Pending Megger test on damaged drum.

8. Material Issue Slip

Field	Detail
Issue Slip Number:	MIS-HEL-26-812
Date:	05-Aug-2026
Requesting Subcontractor:	Electra Erectors LLC
WBS / Cost Code:	WBS-3.2 (AC Cable Pulling Block A)
Item Description:	4C x 240 Sq.mm Al Armoured Cable
Quantity Requested:	2,500 meters
Quantity Issued:	2,500 meters (Drums #AC-44, AC-45, AC-46)
Authorized By:	<i>[Signed Construction Manager]</i>
Received By:	<i>[Signed Subcontractor Rep]</i>

9. Shortage / Damage Report

Field	Detail
Report Number:	SDR-HEL-26-015
Related MRN:	MRN-HEL-26-405
Item Description:	33kV Cable Termination Kits
Quantity Expected:	60 Units
Quantity Received:	56 Units

Variance:	-4 Units (Shortage)
Action Taken:	Payment hold initiated for variance value. Supplier notified to air-freight shortage within 5 days.

10. Subcontract Work Order Key Sections

Section	Description
Work Order No:	WO-HEL-CIV-005
Reference MSA:	Master Service Agreement MSA-2025-09
Subcontractor:	Tashkent Earthworks Co.
Scope of Work:	Excavation and backfilling of 5km main 33kV cable trench as per drawing AFC-TR-01.
Unit Rates:	\$4.50 per cubic meter of excavation.
Measurement:	Joint measurement sheet signed weekly by EPC Site Engineer.
HSE Requirements:	Hard barricading required. Daily toolbox talks mandatory.

SCM Interfaces

The SCM department does not operate in a vacuum; its success relies entirely on seamless cross-functional interfaces.

With Engineering, SCM translates static technical designs into dynamic procurement actions. Engineering provides the foundational MTOs and precise data sheets. When vendors submit their engineering drawings post-order, SCM routes these submittals to the engineering team for approval, tightly managing the turnaround time to prevent manufacturing delays. Furthermore, engineering personnel are required to attend FATs alongside SCM to validate complex electrical performance metrics.

With Planning, SCM continuously synchronizes the procurement schedule against the baseline construction schedule. Through the weekly Expediting Report, SCM surfaces delivery forecasts. If a transformer delivery slips by three weeks, the planner inputs this delay into Primavera P6. If the delay consumes the project float and pushes the critical path, SCM and Planning collaboratively develop mitigation strategies, such as re-sequencing the civil foundation works to minimize idle time on site.

With Finance, SCM manages the project's lifeblood: cash flow. SCM negotiates the payment terms, but Finance executes them. SCM initiates the Letter of Credit (LC) process, defining the required shipping documents and presentation periods, while Finance liaises with the Central Bank of Uzbekistan and issuing banks to establish the LC⁵. SCM also tracks the validity of Advance Payment Bank Guarantees, ensuring they do not expire before the equipment is delivered.

With Contracts, SCM manages the legal periphery of vendor and subcontractor relationships. If a vendor damages site property or delivers late, SCM coordinates with the Contracts team to issue formal notices of Liquidated Damages or back-charges. Conversely, if engineering issues a design change after a PO is signed, SCM and Contracts negotiate the Variation Order to compensate the vendor fairly while protecting the project budget.

With Quality, SCM ensures that what was purchased is actually what arrives. The Quality department conducts the incoming material inspections based on the MRNs generated by SCM. Quality verifies Mill Test Certificates and orchestrates any necessary third-party laboratory testing. If material fails, SCM relies on the Quality department's Non-Conformance Report to enforce warranty replacements from the vendor.

Common SCM Problems and Resolutions

The reality of utility-scale EPC execution involves constant friction. SCM is defined by how it resolves these inevitable crises.

When a multi-megawatt module shipment is held at customs for three weeks, SCM must immediately diagnose the administrative blockage. In Uzbekistan, this is frequently caused by a mismatch between the GTD declaration and the EEISVO contract registration, or missing Certificates of Conformity. SCM must mobilize its licensed customs broker to physically interface with the authorities, update the Single Window portal, and simultaneously warn the site planning team to reallocate module installation crews to structural erection to prevent idle labor burn.

If a 220kV transformer is delivered with suspected transport damage, the process is highly regimented. The 3-axis impact recorder data is immediately downloaded²⁹. If the G-force exceeds the 4g safety threshold³³, the equipment is quarantined on its trailer. SCM instantly notifies the vendor, the project owner, and the marine cargo insurance provider to initiate a claim under the Institute Cargo Clauses (A)³¹. The vendor is mandated to dispatch field service engineers to conduct an internal inspection and an SFRA test to confirm core integrity before the EPC assumes the risk of unloading.

When a subcontractor claims additional payment for scope allegedly outside the Work Order, SCM coordinates a strict contractual defense. SCM requests the subcontractor's formal variation notice, ensuring it complies with the time-bar provisions of the contract. SCM then

cross-references the claim against the approved AFC drawings and the original BOQ. If the work was truly an unrecorded engineering addition, SCM processes a Variation Order; if it was within the original scope, the claim is formally rejected in writing to prevent arbitration risks. If a critical 33kV cable quantity is discovered to be 20% short of the MTO estimate during the final stages of pulling, SCM initiates an emergency procurement protocol. Because international ordering via LC and rail freight would take months, SCM bypasses the standard process. They search local Central Asian markets or tap into the corporate network to cannibalize stock from other ongoing projects. If local sourcing fails, SCM authorizes expedited air freight for the missing length, simultaneously working with engineering to approve straight-through splice kits to utilize any off-cut scrap pieces on site. When a critical vendor fails a Factory Acceptance Test—for instance, a GIS system fails its dielectric withstand test—SCM immediately halts the dispatch milestone. SCM forces the vendor into a Root Cause Analysis protocol. The expeditor works with the factory to compress the rework schedule, attempting to recover the lost time by authorizing weekend shifts. Crucially, SCM formally notifies the vendor that the delivery delay caused by the failed FAT will trigger Liquidated Damages, ensuring the EPC is financially insulated from the manufacturer's quality failure.

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